DAY 25 | POST-READ

SLMs and Token Optimisation: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
SLMs	Small, usually open-weight models that run locally and do narrow tasks well.
SLM or LLM	Decide by how hard, how often, how private and how fast.
Quantisation	Fewer bits per weight: FP16 2 bytes, INT8 1 byte, INT4 half a byte.
LoRA	Train small adapter matrices instead of the whole model; use after prompting and RAG.
Token prices	Input, cached input, output and reasoning tokens are priced differently.
Optimisation	Caching, trimming, top-k, output limits, batching, reasoning effort, routing.
FinOps	Budgets, per-user quotas and a daily usage check, for app calls and for Codex.

Part B: Going beyond today

1. Serving SLMs in production

In production, small models are served by inference servers that batch requests and keep the model in GPU memory. You then call them over HTTP, much like an API LLM, but you run and patch the server yourself.

2. Distillation

A large model can label thousands of examples, and a small model is then trained on those labels. The small model learns the narrow task well and costs far less to run. Check the provider's terms before doing this.

3. Semantic caching

Prompt caching reuses an identical prefix. Semantic caching goes further: if a new question means the same as an old one, return the stored answer. It saves money but can return stale answers, so set an expiry.

4. Measuring the escalation rate

Routing only saves money while most requests stay on the small model. Track the share that escalates to the LLM every day. A sudden rise often means new ticket types or drift (Day 35).

5. Cost in the quality gate

On Day 32 you add cost per query to the CI quality gate, so a change that doubles cost fails like a change that breaks accuracy.

Part C: Check yourself

- Give two reasons a telecom operator might run an SLM on-prem.
- How much memory do the weights of a 3B model need at INT4?
- Why should stable text come first in a prompt?
- What happens to routing savings if the escalation rate doubles?
- What should a daily usage check look for?

Part D: Practice ideas for tonight

- Add five tricky tickets to the comparison and see which model fails first.
- Change the cached share in the calculator from 0.5 to 0.8 and record the saving.
- Write a one-line budget rule for your team's Codex use.

Coming up on Day 26: LLM orchestration, tool calling and LangGraph.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today